

KEY

Science 24

UNIT D WORKBOOK



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Chapter 13: A Need For Safety

Define the following:

Statistics

numerical facts gathered and analyzed in an effort to identify patterns (p. 234)

Reaction Time

time it takes to recognize a situation and act on it (p. 237)

Distraction

anything that takes attention away from a task (p. 237)

Stimulant

substance that increases heart rate and speeds up reaction time; at high doses, can impair reaction time because it makes it more difficult for the person to focus (p. 240)

Impaired

unable to function normally (p. 240)

Depressant

substance that slows reactions, lowers co-ordination, and increases drowsiness (p. 240)

BAC

Blood Alcohol Concentration; measure of the amount of alcohol in the bloodstream; usually measured in milligrams of alcohol per millilitre of blood (p. 241)

Drink

practical guide for monitoring alcohol consumption that defines a standard drink as either 45 mL of hard liquor (40% alcohol), 150 mL of wine (11% alcohol), or 350 mL of beer (4% alcohol) (p. 241)

Impaired driving

legal term for the offence of driving with a blood alcohol concentration of more than 0.08; may be applied to other forms of impairment (p. 241)

Trauma

physical and emotional stress or injury resulting from an accident (p. 243)

Whiplash

injury caused by muscles and ligaments of the neck being stretched or compressed too much (p. 243)

Grief

feelings of regret, sadness, or loss (p. 244)

Therapy

medical treatment that does not involve surgery or drugs (p. 244)

Maintenance

continual care that ensures proper functioning (p. 245)

1. In Canada how much more likely will you die in a road collision than on an airplane? (pg. 234)

50 times more likely to die in a road collision than you are to die in an airplane

2. What type of injuries do you get in a car crash? (pg. 235)

- **cuts, bruises, and broken bones;**
- **whiplash;**
- **serious burns;**
- **internal injuries; and**
- **head and spinal injuries may lead to blindness, paralysis, or brain damage.**

3. Even if alert what are some factors that might affect your ability to stop? (pg. 237)

The road may have been icy.

Your car might have had bald tires or poor brakes.

Some drivers have a habit of following too closely.

4. Why shouldn't you follow too closely? (pg. 237)

You do not always have enough room to stop if the car in front stops suddenly

5. List 4 things that slow your reaction time. (pgs. 237, 240, 241)

- **fatigue**
- **distractions**
- **alcohol**
- **prescription and non-prescription drugs**
- **illness**

6. Are cell phones a problem when driving? Why or why not? (pg. 237)

Statistics suggest they are.

Dialing or having a conversation on a cellular phone can slow reaction time by up to three times.

7. What should you do if you feel fatigued when driving? (pg. 240)

Pull over to a safe parking place and take a nap.

8. Why are stimulants dangerous to take when driving? (pg. 240)

There are three problems with this strategy:

- **the effect is short-lived;**
- **it is possible to over-stimulate your system and negate the effects;**
- **when the effect wears off, individuals are less alert than they were originally.**

9. Why does Alcohol slow reaction time? (pg. 240)

Alcohol slows reaction time because it reduces activity in the central nervous system, resulting in a loss of fine motor co-ordination.

10. What is the legal blood alcohol limit? (pg. 241)

0.05 (0.05 is the new legal limit...0.08 is an old standard)

11. What factors affect your BAC? (pg. 241)

Size/body weight of the individual drinking

Amount of food eaten

Amount of sleep/tiredness

Rate of alcohol consumption

12. Put the following people in order from drunkest to Sober.

a. A 300 lb male drinks 4 drinks in 2 hours. **3**

b. A 90 lb female who has 2 drinks in 1 hour. **1**

c. A 200 lb 55-year-old alcoholic who has 3 drinks in 2 hours. **4**

d. A 90 lb female who has 2 drinks in 1 hour on a full stomach. **2**

13. Use the following chart for the next question.

BAC Based on Number of Drinks

Body Mass Max. (kg)	Female Number of Drinks						Male Number of Drinks					
	1	2	3	4	5	6	1	2	3	4	5	6
45	0.05	0.10	0.152	0.203	0.253	0.304	0.043	0.087	0.130	0.174	0.217	0.261
57	0.04	0.080	0.120	0.162	0.202	0.244	0.034	0.069	0.103	0.139	0.173	0.209
68	0.034	0.068	0.101	0.135	0.169	0.203	0.029	0.058	0.087	0.116	0.145	0.174
79	0.029	0.058	0.087	0.117	0.146	0.175	0.025	0.050	0.075	0.100	0.125	0.150
91	0.026	0.050	0.076	0.101	0.126	0.152	0.022	0.043	0.065	0.087	0.108	0.130
102	0.022	0.045	0.068	0.091	0.113	0.136	0.019	0.039	0.058	0.078	0.097	0.117
114	0.020	0.041	0.061	0.082	0.101	0.122	0.017	0.035	0.052	0.070	0.087	0.105

- a. The body eliminates 0.015BAC units per hour. The legal limit is 0.05. If you drank 4 drinks in 1 hr how long will you have to wait before you can drive. (pg. 228)

(Your BAC – 0.050 BAC) = ANSWERS WILL VARY DEPENDING ON
0.015 BAC/h YOUR SEX AND BODY MASS

- b. How long would it take a 68kg Female to sober up after 6 drinks in an hour?

$$\frac{(0.203 \text{ BAC} - 0.050 \text{ BAC})}{0.015 \text{ BAC/h}} = 10.2 \text{ h}$$

14. What part of the car will prevent whiplash? (pg. 243)

headrest

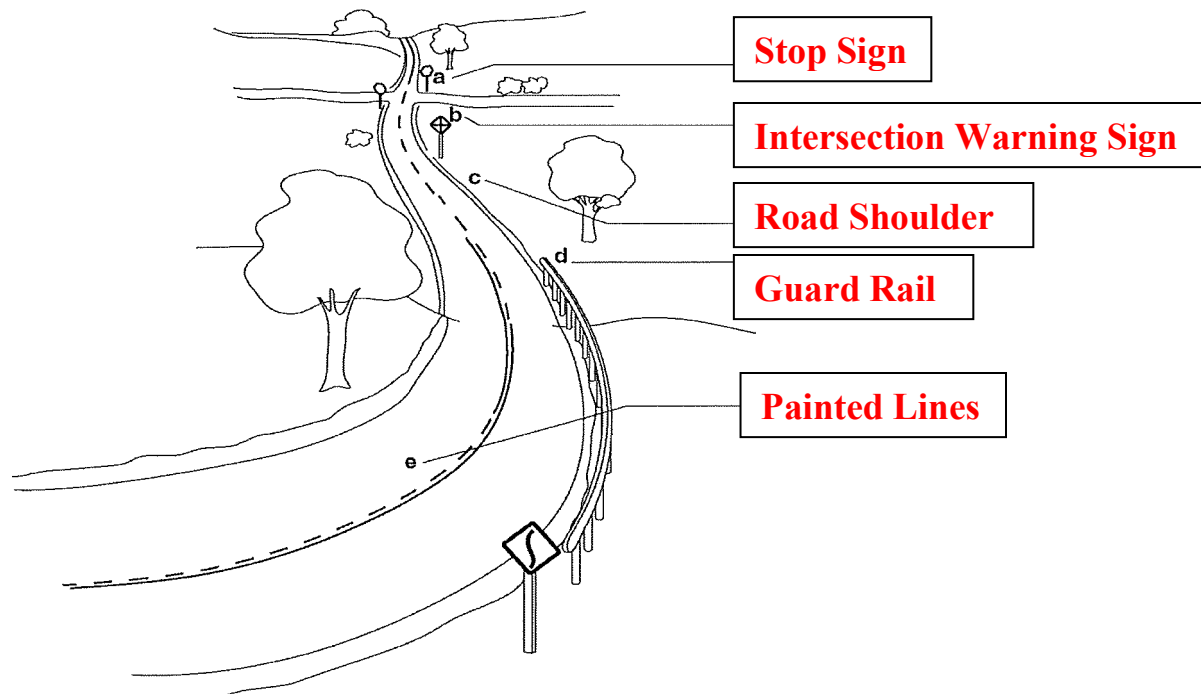
15. List 4 safety features that your car has. (pg. 245)

headrests, seat belts, air bags, anti-lock braking systems (ABS), crumple zones, side impact beams with steel frames, safety glass

16. Why should you perform regular maintenance on a vehicle? (pg. 245)

Maintenance or continual care makes sure that safety features work properly and continue to protect you

17. Label the safety features on this road. (pg. 247)



Chapter 14: The Nature of Speed

Define the following:

Distance

change in position (p. 252)

Time

interval between events (p. 252)

Velocity

speed; distance travelled during a specific time interval (p. 252)

Distance/time graphs

graphic representations of speed (velocity) (p. 256)

Variables

aspect of an investigation that changes because it is manipulated or because it responds to manipulation;

see also *manipulated variable* and *responding variable* (p. 256)

Manipulated variable

aspect of an investigation that the researcher changes;

see also *variable* and *responding variable* (p. 256)

Responding variable

aspect of an investigation that the researcher measures as it changes;

see also *variable* and *manipulated variable* (p. 256)

Run

difference in horizontal position on a graph; shows the time interval between two points on a distance/time graph (p. 260)

Rise

difference in vertical position on a graph; shows the distance interval between two points on a distance/time graph (p. 260)

Slope

steepness of the line on a graph; measured by dividing rise by run (p. 260)

Braking distance

**distance a vehicle travels between when the driver applies the brakes and when the vehicle comes to a stop;
see also stopping distance (p. 254)**

Stopping distance

**distance that a vehicle travels during reaction time (until the vehicle stops);
see also braking distance (p. 264)**

Following distance

Distance (measured in seconds) at which it is considered safe to follow another vehicle under good driving conditions; generally accepted to be two seconds (p. 265)

1. What is another word for velocity? (pg. 252)

Speed

2. What is the formula for calculating velocity? (pg. 254)

$$\text{velocity} = \frac{\text{distance}}{\text{time}} \quad \text{OR} \quad v = \frac{d}{t}$$

3. Rearrange the formula above to solve for distance. (pg. 254)

$$t \ v = \frac{d}{\cancel{t}} \quad d = vt$$

4. How does increasing the distance affect the time if the velocity stays the same? (pg. 254)

If velocity stays the same, increasing the distance will increase the time.

5. How would doubling the velocity while keeping distance the same affect time of travel? (pg. 254)

Doubling the velocity for the same distance would cut the time in half.

6. Cameron drives 300 km in 2 hours. How fast is he traveling? Show your work. (pg. 254)

$$\begin{array}{l} d = 300 \text{ km} \\ t = 2 \text{ h} \end{array} \quad v = \frac{d}{t} = \frac{300 \text{ km}}{2 \text{ h}} = 150 \text{ km/h}$$

7. Brandon is running to Macs to grab a slurpee at 10 minute break (600s). Round trip to Macs from NSO is 1600m. Can Brandon make it in time if he runs 10m/s?

$$\begin{array}{l} v = 10 \text{ m/s} \\ t = 600 \text{ s} \end{array} \quad d = vt = 10 \text{ m/s} \times 600 \text{ s} = 6000 \text{ m}$$
$$d = 1600 \text{ m} \quad t = \frac{d}{v} = \frac{1600 \text{ m}}{10 \text{ m/s}} = 160 \text{ s}$$

****Brandon will make it in time!**

8. On a distance vs time graph on what axis does the time go? (pg. 256)

Time goes on the x-axis - horizontal

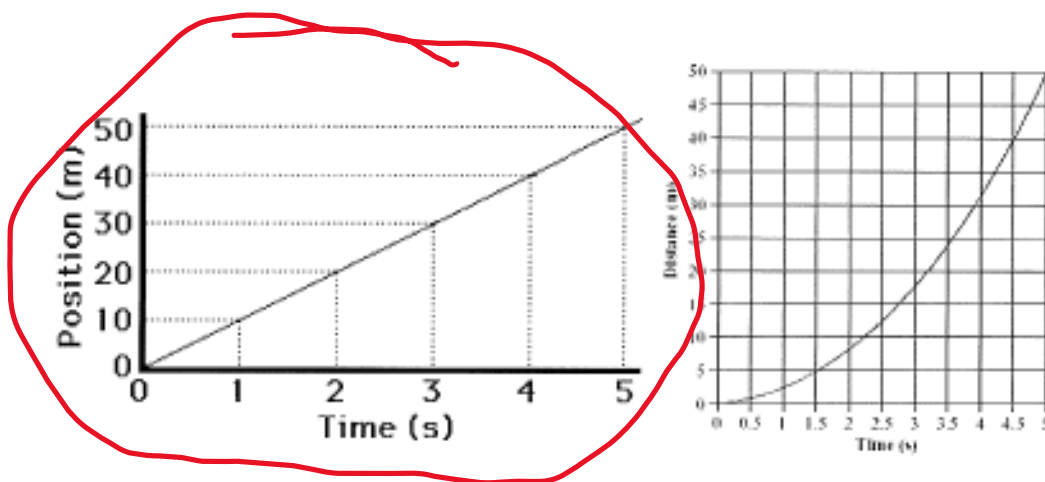
9. On a distance vs time graph what axis does the distance go? (pg. 256)

Distance goes on the y-axis - vertical

10. What type of line do we draw through the points on a distance vs time graph? (pg. 256)

Line of best fit

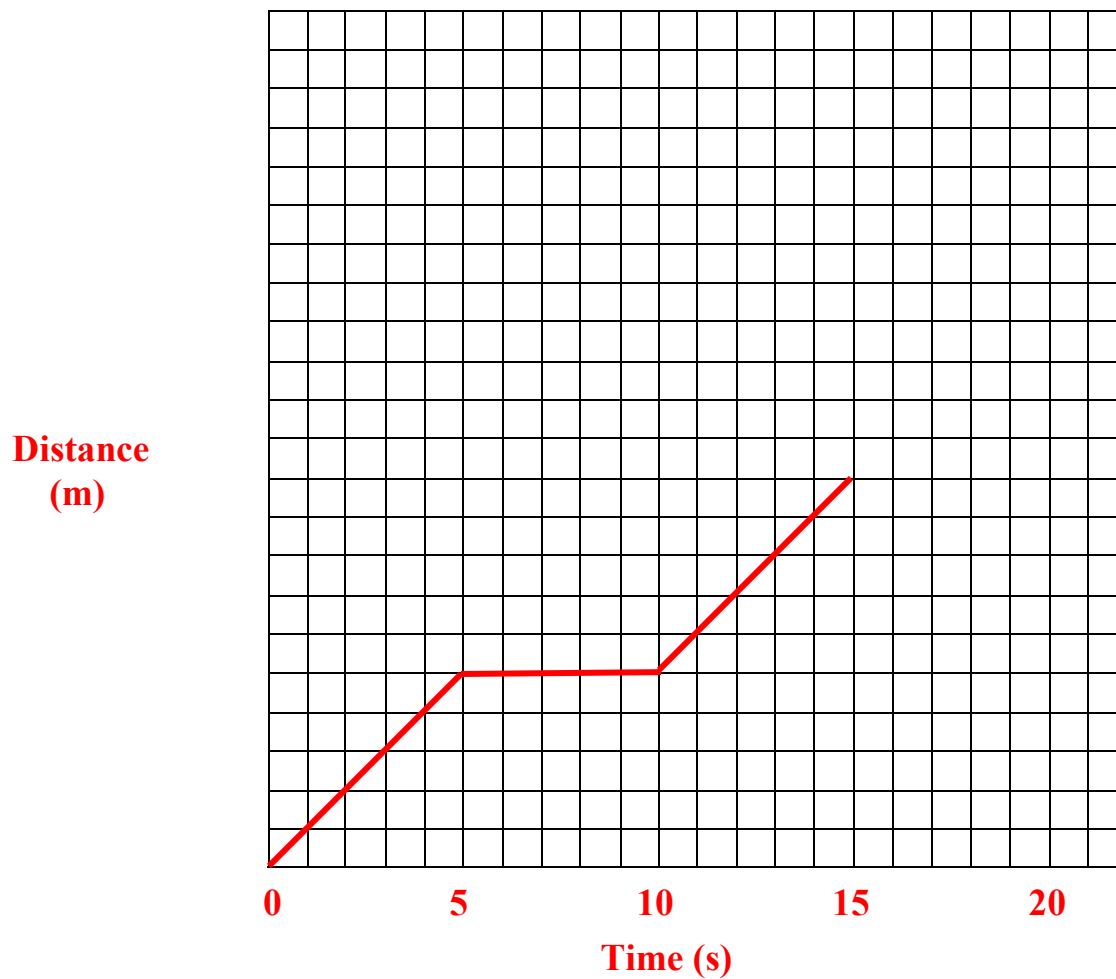
11. **Circle** the following graph that show someone traveling at a constant speed? (pg. 257)



Straight line!!!

12. Draw a graph rough of the following situation. Tyler runs at a constant speed for 5 seconds. He then stops for 5 seconds and runs again.

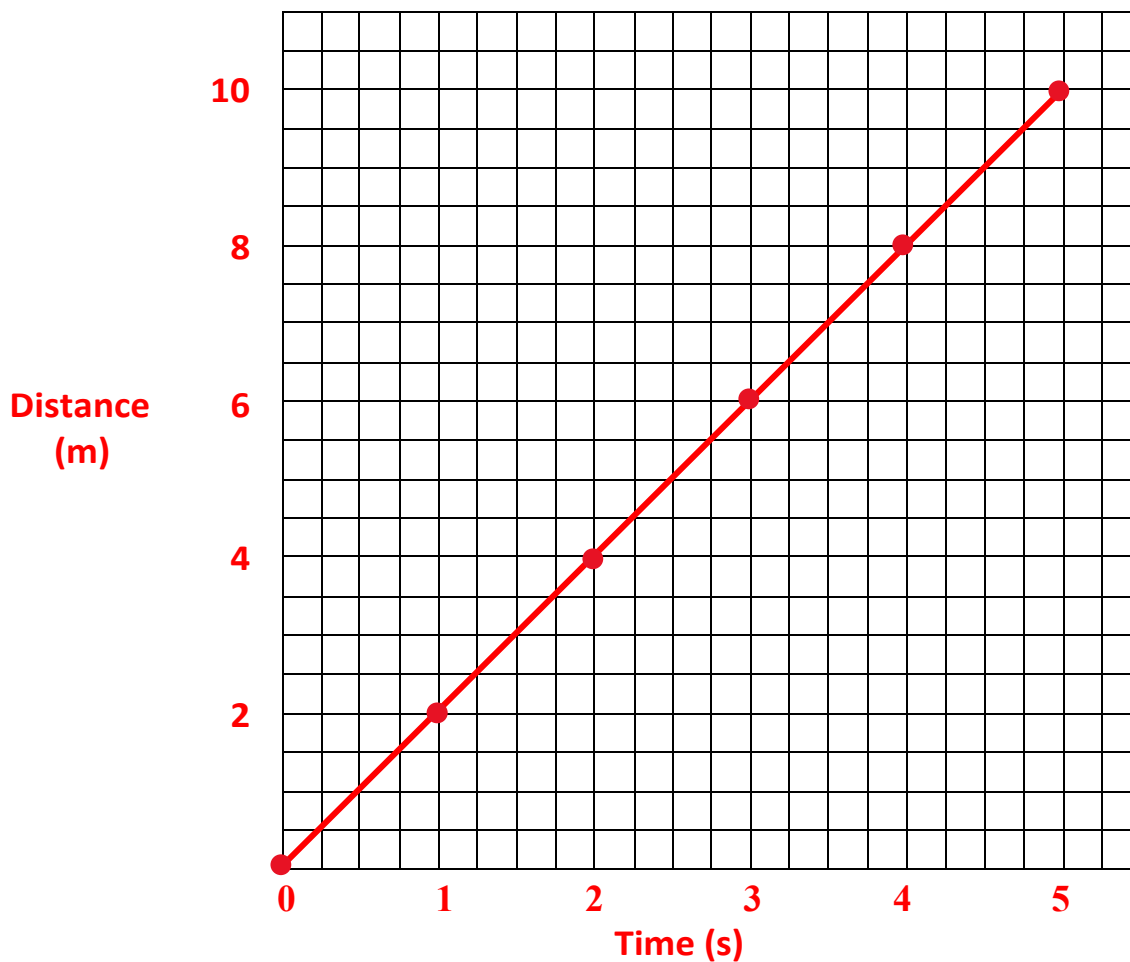
Distance – Time Graph



13. Graph the following table of values on a graph. Be sure to create a title and to label all your axis.

Time	0 s	1 s	2 s	3 s	4 s	5 s
Distance	0 m	2 m	4 m	6 m	8 m	10 m

Distance – Time Graph



14. Calculate the slope of the object. (pg. 260)

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{d}{t} = \frac{10 \text{ m} - 0 \text{ m}}{5 \text{ s} - 0 \text{ s}} = 2 \text{ m/s}$$

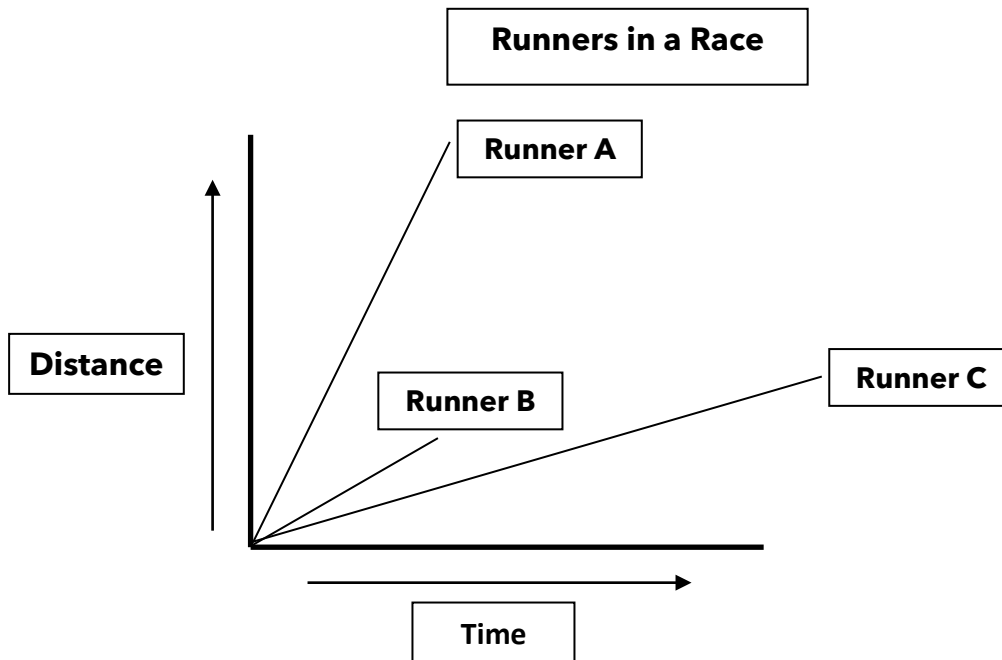
15. What does the slope give us? (pg. 261)

speed/ velocity

16. What does the horizontal line a distance vs. time graph represent?

The horizontal line (run) on a distance/time graph (x-axis) represents time.

17. Use the following graph to answer the next 3 questions.



a) Which runner was running the fastest? **Runner A**

b) Which runner went the shortest distance? **Runner B**

c) Which runner ran for the longest time? **Runner C**

18. Complete the following Chart.

Trip	Distance	Time	Speed
Edmonton to Red Deer	148.5 km	1.35 h	110 km/h
Jasper to Hinton	66 km	0.66 h	100 km/h
Morinville to Vegas	1966 km	16.38 h	120 km/h
Morinville to Drumheller	275 km	2 h	137.5 km/h
Morinville to Jasper	311 km	3.11 h	100 km/h
Morinville to Edmonton	28 km	0.167 h (10 min)	167.6 km/h

19. What is the difference between Braking distance and Stopping distance? (pg. 264)

Braking Distance

- The distance that a vehicle travels after the driver applies the brakes.

Stopping Distance

- includes the distance travelled during reaction time plus the braking distance.

20. When should you increase following distance? (pg. 265)

- At higher speeds
- When road conditions are not ideal (wet, icy, gravel etc)

Chapter 15: The Physics of Collisions

Define the following:

Momentum

measure of an object's motion and likelihood that the object will remain in motion (p. 270)

Force

any push or pull on an object (p. 270)

Braking

force applied by brakes to stop a vehicle (p. 272)

Friction

force resisting motion, caused by substances rubbing together; rubbing (p. 272)

Impulse

force that changes momentum; force applied over time (p. 274)

Law of Conservation of Momentum

total momentum of colliding objects stays the same after the collision (p. 280)

Total momentum

**sum of momentum of colliding objects;
see Conservation of Momentum, Law of (p. 281)**

1. When will an object that is not moving begin to move? (pg. 270)

Objects that are not moving will not move until they are pushed or pulled with enough force.

2. When will an object that is moving stop? (pg. 270)

Objects that are moving will keep moving until enough force is added in an opposite direction to stop them.

3. How do we calculate momentum? (pg. 271)

Momentum is calculated by multiplying the mass (kg) of an object times the velocity (m/s) it is travelling.

$$\begin{array}{ccccc} \text{momentum} & = & \text{mass} & \times & \text{velocity} \\ (\text{kg}\bullet\text{m/s}) & & (\text{kg}) & & (\text{m/s}) \end{array}$$

4. Fill in the following chart. (pg. 271)

Object	Mass	Velocity	Momentum
Compact Car	750 kg	100 km/h (30 m/s)	22500 kg•m/s
Adult female on a bike	70 kg	30 km/h (9 m/s)	630 kg•m/s
Adult female	70 kg	5 km/h (1.5 m/s)	105 kg•m/s
Transport truck	2500 kg	50 km/h (15 m/s)	37500 kg•m/s
Transport truck	2500 kg	100 km/h (30 m/s)	75000 kg•m/s

5. Give an example of an object with a very high momentum and one with a very low momentum. (pg. 271)

- **Slugs have a relatively low momentum.**
They are relatively light and do not move fast.
- **Airplanes have a huge momentum.**
They have a mass of many thousands of kilograms and travel at several hundred kilometres per hour.

6. How is momentum affected in each situation. (pg. 271)

a. Doubling velocity

doubling the velocity doubles the momentum

b. Doubling mass

doubling the mass doubles the momentum

c. Doubling both velocity and mass

doubling both the velocity *and* mass quadruples the momentum

7. How do brakes work? (pg. 272)

Brakes apply a force of friction that stops vehicles by decreasing their momentum.

Applying the brake pedal activates machinery that presses brake pads against the inside of brake drums on the vehicle's wheels.

8. Why don't brakes work as well in wet or icy conditions? (pg. 272)

roads that are slippery due to rain, ice, or snow will reduce the amount of friction and therefore slow the speed of braking.

9. How do we calculate impulse? (pg. 274)

The combination of the size of the force and the time during which the force acts is called impulse.

$\text{impulse} = \text{force} \times \text{time}$

Impulses change momentum — either increasing or decreasing it.

The bigger the impulse is, the bigger the change in momentum will be.

10. Solve the following questions:

a. A football player collides with another player. The force of the collision is 100N. The collision lasts 2 seconds. What is the impulse?

$\text{impulse} = \text{force} \times \text{time} = 100 \text{ N} \times 2 \text{ s} = 400 \text{ N}\cdot\text{s} = 400 \text{ kg}\cdot\text{m/s}$

b. A baseball player feels a frictional force of 400N as she slides into home. What is the impulse if the slide lasts 0.5 seconds?

$\text{impulse} = \text{force} \times \text{time} = 400 \text{ N} \times 0.5 \text{ s} = 200 \text{ N}\cdot\text{s} = 200 \text{ kg}\cdot\text{m/s}$

11. How does increasing your stop time protect you from injury? (pg. 274)

By increasing the time, you decrease the force necessary to create the same change in momentum.

12. Is the force needed to stop a car the same whether you take a long time or a short time to stop? Explain.

No, the force needed to stop a vehicle is different depending on whether you take a long or short time to stop.

The longer the time, the less force will be required.

The shorter the time, more force will be required to stop.

13. In a collision that stops a vehicle, how does the original velocity relate to the force experienced by vehicle and passenger? In other words, if a car hits a brick wall how does the speed at which it hits the wall affect the force felt by the passenger? (pg. 277)

The force felt by a passenger is directly correlated to the velocity.

The larger the velocity, the more force is felt.

14. Use the following diagrams and determine what happens to momentum. (pg. 281)

a. Two cars of the same size and traveling at the same speed hit in a head on collision.

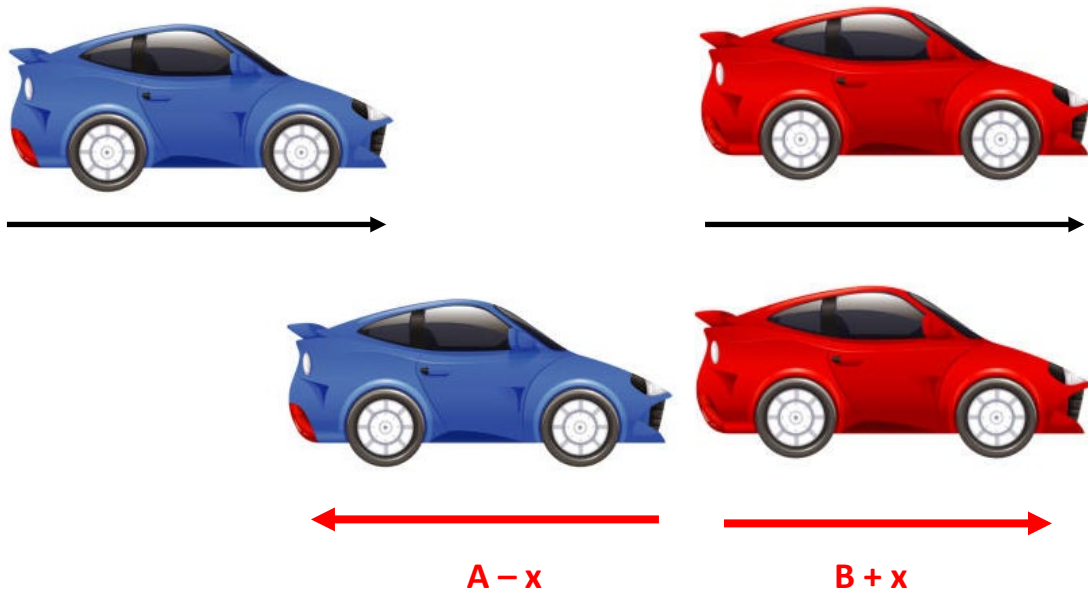


Equal and opposite momentums cancel each other.

The cars will stop.

b. Draw the arrows and explain what would happen to momentum after collision.

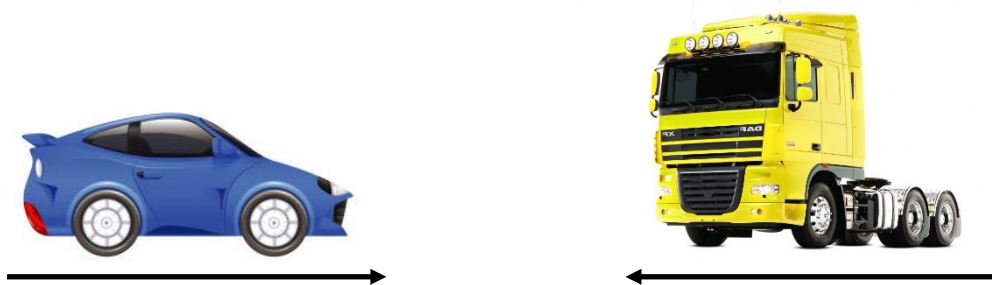
Two cars traveling in the same direction hit.



**The vehicle in front will speed up, and the one behind will slow down.
The momentum from the car in the rear is transferred to the car in front.**

c. What would happen in a head on collision between these two vehicles?

Assuming both vehicles were traveling at the same speed.



**The transport truck (having much greater momentum) will slow down,
but maintain movement in that direction, pushing the car (whose momentum
and direction of travel will completely reverse).
Momentum of the truck is lessened, but that of the car is completely reversed.
Momentum lost from the truck goes into changing the car's direction of travel.**

KEY

Chapter 16: The Technology of Safety

Define the following:

Crash test dummies

Manikins designed to behave like the human body during a collision; fitted with sensors to record data (p. 288)

Crumple zones

parts of a vehicle designed to crush when hit, absorbing much of the energy in a collision; include hood, trunk, and fenders (p. 288)

Restraining features

hold occupant in place; prevent occupant from hitting the dashboard, windshield, sides, or roof of the car, or from being thrown out of the vehicle during a collision; include seat belts, air bags, headrests, and child seats (p. 289)

Operational features

parts of a vehicle that operate all the time to keep occupants safe; include headlights, rear-view and side-view mirrors, tail and signal lights, brakes, etc. (p. 289)

Structural features

parts of a vehicle that protect the people inside; include crumple zones and rigid features such as heavy frame, roll cage, and side-impact beams that will not crumple or break under extreme pressure; see crumple zones (p. 289)

Road safety features

traits of a road that are designed to decrease the force of a collision by slowing the impulse; see crash cushion and guardrails (p. 291)

Crash cushion

road feature designed to decrease the force of a collision by slowing the impulse; barrels of water or sand; see also road safety features and guardrails (p. 291)

Guardrails

road features designed to keep vehicles on the road and to decrease the force of a collision by slowing the impulse and thus absorbing the force of the collision; see crash cushion and road safety features (p. 291)

2-point seat belts

see lap belts (p. 292)

Lap belts

2-point seat belts; vehicle seat belts designed to hold the occupant in place by securing the person to the seat with a strap placed across the pelvis (p. 292)

1. What do safety features do on a car? (pg. 288)

The safety features on vehicles work to reduce the force of a crash.

You learned that the force resulting from a collision is smaller if the change in momentum occurs over a longer period of time.

Safety features serve to slow down the collision. As a result, the occupants of the car experience less force and, therefore, receive fewer injuries.

2. Why do vehicles have crumple zones? (pg. 288)

The crumple zones slow down the collision felt by the occupants and increase the time of the collision between the occupants and the interior of the vehicle, thus reducing the force experienced by the occupants.

This reduces the damage experienced by the occupants of the vehicle.

3. List 4 examples of a restraining feature. (pg. 289)

seat belts, air bags, headrests, and child seats

4. Why are your rear and side mirrors an example of an operational feature? (pg. 289)

Operational features operate all the time to keep you safe while driving.

Rear- and side-view mirrors assist the driver in seeing all parts of the road.

5. What are the two types of structural features? List an example of each. (pg. 289)

– Rigid features include the heavy frame, a roll cage, and side-impact beams in the doors. Each of these is built from materials that will not crumple or break under extreme pressure. In this way, they are designed to protect the people inside from anything that may hit the car.

– Crumple zones include the hood, trunk, and fenders. These may be completely crushed in a bad accident. If the rigid frame stays intact, however, the cabin of the vehicle will keep the occupants safe.

6. Why are the rigid features of a car surrounding the cabin of the vehicle? (pg. 289)

Rigid features are built from materials that will not crumple or break under extreme pressure.

In this way, they are designed to protect the people inside from anything that may hit the car.

7. What is a major cause of injury in a car with no crumple zones? Why? (pg. 290)

Hit the hard dashboard or the rigid steering wheel.

The force of impact was not absorbed by a crumple zone causing the occupant to continue forward and hit the dashboard or steering wheel with a large force.

8. List 4 road features that were designed to reduce the force of impact. (pg. 291)

Crash barrier/cushion (sandbags, water barrels) , guardrails, break away light poles

9. What type of injury would happen if 2-point belts were in the front seat? (pg. 292)

The 2-point lap belt would not prevent the upper body from continuing to move forward and the upper body would strike the dashboard or steering wheel

10. Engineers have not found a way to restrain the head during a collision.

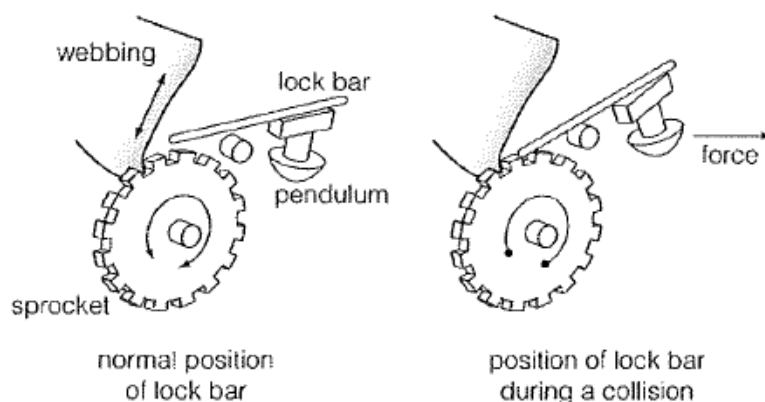
What safety features are designed to protect the head? (pg. 292)

a raised headrest along with the seat belt

11. Why does the webbing of a seat belt stretch? (pg. 293)

stretching allows the passenger to stop more gradually and feel a smaller force

12. Explain how this structure works. (pg. 293)



If the vehicle experiences a sudden stop, the pendulum in the ratchet mechanism swings forward.

This locks a bar in place that prevents more webbing from coming out.

13. Circle the right answer:

Babies heads are (**larger**/ smaller) in proportion to their body. (pg. 296)

14. Why should car seats be facing the rear of the vehicle? (pg. 297)

The force of a crash would be absorbed by the entire back of the child.

15. What type of gas does an air bag fill up with? (pg. 298)

Nitrogen gas – $N_{2(g)}$

16. What was the problem with the first-generation air bag? (pg. 298)

First generation air bags deployed so fast that they hit the driver at over 300 km/h

17. List 2 features of the third-generation air bags. (pg. 301)

There are now seat sensors that detect the mass of the occupant and adjust the force of the air bag accordingly.

Some of the new air bags also have switches so they can be turned off if there is a child in the front seat.